

July 5, 2026 – Firmware Architecture

Today I began developing firmware for the ESP32 telemetry platform using C++ and PlatformIO. I organized the project into separate header (.h) and source (.cpp) files to create a modular firmware architecture that simplifies development and future expansion.

I implemented a TelemetryData structure to store battery voltage, motor temperature, wheel RPM, vehicle speed, pitch, and roll. To verify the software architecture before integrating hardware, I developed simulated telemetry values and displayed them through the Serial Monitor.

Finally, I initialized the GitHub repository and integrated the firmware into version control to document development and track future changes.

Next Steps

- Replace simulated telemetry with live sensor data.
- Integrate battery voltage, temperature, Hall-effect, and IMU sensors.
- Implement Bluetooth Low Energy (BLE) communication.
- Begin development of the mobile telemetry application.